

AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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RESEARCH SUMMARY

State Engineer (ESI-SBA) with a focus on **Structural Generalization** and **Sequence Modeling**. Expert in transformer alternatives (**Mamba-SSM**), Graph Neural Networks, and algorithmic optimization. Strong background in mathematical foundations of Deep Learning.

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

RESEARCH PROJECTS

Generative Surrogate Architectures (Mamba-SSM) 2026
Research on structural generalization for sequence-to-sequence modeling in industrial CPS.

- Developed a generative simulator using **Mamba-SSM** to handle long-range dependencies.
- Implemented **xLSTM** with C++ JIT for high-speed structural inference.
- Investigated **Koopman Theory** for linearizing non-linear structural dynamics.

Career Connect — Structural GNN Intelligence 2024 – 2025
Architected **Graph Neural Networks** for semantic structural matching.

- Researched heterogeneous relation embeddings and network centrality.
- Focused on generalization across diverse relational structures.

NLP Optimization — DistilBERT & Metaheuristics 2025
Hyperparameter optimization for transformer-based LLMs using PSO and GA.
Geo-RAG — Semantic Knowledge Retrieval 2025
RAG system for multi-source knowledge extraction and structural reasoning.

TECHNICAL SKILLS

- Sequence Modeling:** Transformers, Mamba-SSM, xLSTM, Informer, TCN.
- Graph AI:** GNN (GraphSAGE, GAT), Structural Centrality, PyG.
- Mathematics:** Linear Algebra, Optimization (PSO, GA), PINN, Koopman Theory.
- Engineering:** PyTorch, C++, Docker, MLflow, CUDA, Linux.

LANGUAGES

Arabic (Native), English (Advanced - Professional Research Level), French (Fluent).